

2.1 Ecosystems of the Planet

A variety of ecosystems are spread across the world and these have a number of interacting components and characteristics. This theme develops an appreciation of a number of these ecosystems, before focusing study on coral reefs and tropical rainforests. Both ecosystems

will be examined in terms of their abiotic and biotic components, processes, cycles and their value to humans. Learners explore the sustainable use and management of these bio-diverse ecosystems.

Section	Key Ideas	Content	Scale
2.1.1	Ecosystems consist of interdependent components.	Ecosystems include abiotic (weather, climate, soil) and biotic (plants, animals, humans) components which are interdependent.	R, L
2.1.2	Ecosystems have distinct distributions and characteristics.	- Overview of the global distribution of polar regions, coral reefs, grasslands, temperate forests, tropical rainforests, and hot deserts. - Overview of the climate, plants and animals within these ecosystems.	G
2.1.3	There are major tropical rainforests in the world.	The location of the tropical rainforests including the Amazon, Central American, Congo River Basin, Madagascan, South East Asian and Australasian.	G
2.1.4	There are major coral reefs in the world.	The location of warm water coral reefs including the Great Barrier Reef, Red Sea Coral Reef, New Caledonia Barrier Reef, the Mesoamerican Barrier Reef, Florida Reef and Andros Coral Reef.	G
2.1.5	Bio-diverse ecosystems are under threat from human activity.	- The processes that operate within tropical rainforests, including nutrient and water cycles. - The process of nutrient cycling that operates within coral reefs. Two case studies, including one tropical rainforest and one coral reef, to cover: - the interdependence of climate, soil, water, plants, animals and humans - their value to humans and to the planet - threats to biodiversity and attempts to mitigate these through sustainable use and management.	G, R, N, L